

Math 2551 True/False Bank

Choose whether the following statements are true or false. If the statement is *always* true, pick true. If the statement is *ever* false, pick false.

1 Vectors, Lines, Planes, Quadrics, Vector-Valued Functions

1. True or False: If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^3$, then $\mathbf{u} \times \mathbf{v} = \mathbf{v} \times \mathbf{u}$.
2. True or False: If $\mathbf{T}(t)$ is the unit tangent to $\mathbf{r}(t)$ and $\mathbf{N}(t)$ is the principal normal vector, then $\mathbf{T}(t) \cdot \mathbf{N}(t) = 0$.
3. True or False: A smooth curve in the plane that never crosses itself can have two distinct tangent lines at a given point.
4. True or False: If a spaceship is in orbit around the Moon with a constant speed of 3750 miles per hour, then it is the case that the acceleration of the spaceship is zero.
5. True or False: The sphere $x^2 + (y - 2)^2 + (z + 1)^2 = 6$ has center $(0, -2, 1)$.
6. True or False: If $\mathbf{v}$ and $\mathbf{w}$ are vectors in $\mathbb{R}^3$, then $(\mathbf{w} \times \mathbf{v}) \cdot \mathbf{w} = 0$.
7. True or False: If a vector-valued function $\mathbf{r}(t)$ is differentiable at $t = 2$ then a direction vector for the line tangent to $\mathbf{r}(t)$ at $t = 2$ is $\mathbf{r}'(2)$.
8. True or False: The curvature of a line is 0 at every point on the line.
9. True or False: If $\mathbf{u}$ and $\mathbf{v}$ are two vectors in $\mathbb{R}^3$, then $\mathbf{u} \cdot \mathbf{v}$ is a vector orthogonal to both $\mathbf{u}$ and $\mathbf{v}$.
10. True or False: There is exactly one possible parameterization for any line in space.
11. True or False: If $\mathbf{u}$ and $\mathbf{v}$ are orthogonal unit vectors in $\mathbb{R}^3$, then $|\mathbf{u} \times \mathbf{v}| = 1$.
12. True or False: The surface in $\mathbb{R}^3$ consisting of the points which solve the equation $x^3 + y^3 - z = 27$ is an example of a quadric surface.
13. True or False: $\|\mathbf{v}\| = |v_1| + |v_2| + |v_3|$, where $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$.
14. True or False: If the speed of a particle is zero, its velocity must be zero.
15. True or False: The points $(1, 0, 1)$ and $(0, -1, 1)$ are the same distance from the origin.
16. True or False: The point $(2, -1, 3)$ lies on the graph of the sphere $(x - 2)^2 + (y + 1)^2 + (z - 3)^2 = 25$.
17. True or False: The equation $Ax + By + Cz + D = 0$ represents a line in space.
18. True or False: Any three points in 3 space determine a unique plane.
19. True or False: Any two distinct lines in 3-space determine a unique plane.
20. True or False: If the graph of $z = f(x, y)$ is a plane, then each cross section is a line.
21. True or False: The length of the sum of two vectors is always strictly larger than the sum of the lengths of the two vectors.
22. True or False: $\mathbf{v}$ and $\mathbf{w}$ are parallel if $\mathbf{v} = \lambda\mathbf{w}$ for some scalar $\lambda \neq 0$.
23. True or False: Any two parallel vectors point in the same direction.
24. True or False: Any two points determine a unique displacement vector.

25. True or False: $2\mathbf{v}$ has twice the magnitude as $\mathbf{v}$.
26. True or False: The vector $\mathbf{v} = \langle \frac{1}{2}, \frac{1}{2} \rangle$ is a unit vector.
27. True or False: The vector $\frac{1}{\sqrt{3}}\mathbf{i} - \frac{1}{\sqrt{3}}\mathbf{j} + \frac{2}{\sqrt{3}}\mathbf{k}$ is a unit vector.
28. True or False: The vectors $2\mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ are parallel.
29. True or False: The only way that $\mathbf{v} \cdot \mathbf{w} = 0$ is if $\mathbf{v} = \mathbf{0}$ or $\mathbf{w} = \mathbf{0}$.
30. True or False: The zero vector $\mathbf{0}$ (with magnitude $\|\mathbf{0}\| = 0$) is perpendicular to all other vectors.
31. True or False: Any plane has only two distinct normal vectors.
32. True or False: Parallel planes share a same normal vector.
33. True or False: Perpendicular planes have perpendicular normal vectors.
34. True or False: An equation of the plane with normal vector $\langle 1, 1, 1 \rangle$ containing the point $(1, 2, 3)$ is $z = x + y$.
35. True or False: For any two vectors $\mathbf{u}$ and $\mathbf{v}$, $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$.
36. True or False: For any vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^3$, $(\mathbf{u} \times \mathbf{v}) \times (\mathbf{v} \times \mathbf{u}) = (\mathbf{v} \times \mathbf{u}) \times (\mathbf{u} \times \mathbf{v})$.

2 Functions of Multiple Variables, Partial Derivatives, Optimization

37. True or False: Suppose we have a function $f(x, y)$ and $\mathbf{u} \in \mathbb{R}^2$ is a unit vector. Then $D_{\mathbf{u}}f(a, b)$ is a vector.
38. True or False: In a table of values for a linear function, the columns must have the same slope as the rows.
39. True or False: If $f(x, y) = (x + y)^{50}$, then all of the 100th order partial derivatives of f will be 0.
40. True or False: The total derivative Df of the function $f(x, y, z, w) = 2xyz + 3\ln(w) + \sin(e^{xw+z})$ is represented by a 1×4 matrix.
41. True or False: Suppose $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 1$ along the y -axis, the x -axis, and the line $y = x$ and $f(0, 0) = 1$. Then f must be continuous at $(0, 0)$.
42. True or False: The domain of the function $f(x, y) = \sqrt{x-2} + y$ is the interval $[2, \infty)$.
43. True or False: Suppose $f(x, y)$ is a differentiable function defined on all of $\mathbb{R}^2$, $f_{xx}(0, -1) = -3$, $f_{xy}(0, -1) = f_{yx}(0, -1) = 4$, and $f_{yy}(0, -1) = -6$. Then $(0, -1)$ is the location of a local maximum of f .
44. True or False: The domain of the function $f(x, y) = \ln(x - 2) + y^2$ is the interval $(2, \infty)$.
45. True or False: The contour lines of the graph of $f(x, y) = 2x + 3y$ are parallel lines.
46. True or False: Suppose that the price P in dollars to purchase a used car is a function of C , its original cost in dollars, and its age A in years, i.e. $P = f(C, A)$. Then the sign of $\frac{\partial P}{\partial C}$ is negative.

47. True or False: Suppose the linear approximation of a function $f(x, y)$ at the point $(2, -1)$ is

$$L(x, y) = 4 + 3(x - 2) + 5(y + 1).$$

Then an equation of the tangent plane to the graph of f at the point $(2, -1, 4)$ is $z = 4 + 3(x - 2) + 5(y + 1)$.

48. True or False: The domain of the function $f(x, y, z) = \frac{1}{x-2} + yz^2$ is $(-\infty, 2) \cup (2, \infty)$.
49. True or False: There does not exist a function $f(x, y)$ which is continuous and has continuous partial derivatives such that $f_x(x, y) = x^2 + 2xy$ and $f_y(x, y) = 3xy + y^2$.
50. True or False: A function of two variables is differentiable at the point (a, b) if the surface $z = f(x, y)$ has a unique tangent plane at the point $(a, b, f(a, b))$.
51. True or False: The two-path test says that if there are two paths through the point (a, b) on which the limit of $f(x, y)$ is L , then

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L.$$

52. True or False: If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is any function of two variables, $f_{xy} = f_{yx}$ at every point (x, y) in the domain of f .
53. True or False: If $z = f(x, y)$ and $x = u(s, t)$, then it must be true that $\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s}$.
54. True or False: A function $f(x, y)$ can be an increasing function of x with y held fixed, and be a decreasing function of y with x held fixed.
55. True or False: The graph of the equation $f(x, y) = 2$ is a plane parallel to the xz -plane.
56. True or False: The cross section of the function $f(x, y) = x + y^2$ in the plane $y = 1$ is a line.
57. True or False: The graphs of $f(x, y) = x^2 + y^2$ and $g(x, y) = 1 - x^2 - y^2$ intersect in a circle.
58. True or False: The contours of the graph of $f(x, y) = y^2 + (x - 2)^2$ are either circles or a single point.
59. True or False: If all the contours for $f(x, y)$ are parallel lines, then the graph of f is a plane.
60. True or False: On a weather map, there can be two isotherms (contour lines) which represent the same temperature but do not intersect.
61. True or False: Any surface that is a graph of a 2-variable function $z = f(x, y)$ can be thought of as a level surface of a function of 3 variables.
62. True or False: Any level surface of a function of 3 variables can be thought of as the graph of a function $z = f(x, y)$.
63. True or False: The level surfaces of the function $f(x, y, z) = x^2 + y^2 + z^2$ are cylinders with axis along the y -axis.
64. True or False: If $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}$ everywhere, then $f(x, y)$ is constant.
65. True or False: There exists a function $f(x, y)$ with $f_x = 2y$ and $f_y = 2x$.
66. True or False: The differential of a function $f(x, y)$ at the point (a, b) is given by the formula $df = f_x(a, b)dx + f_y(a, b)dy$. The equation of the tangent plane to $z = f(x, y)$ at the point (a, b) can be used to calculate values of df from dx and dy .

67. True or False: The differential of a function $f(x, y)$ at the point (a, b) is given by the formula $df = f_x(a, b)dx + f_y(a, b)dy$. If dx and dy represent small changes in x and y in moving away from the point (a, b) , then df approximates the change in f .
68. True or False: The function $f(x, y)$ has gradient ∇f at the point (a, b) . The vector ∇f is perpendicular to the level curve $f(x, y) = f(a, b)$.
69. True or False: The function $f(x, y)$ has gradient ∇f at the point (a, b) . The vector ∇f is perpendicular to the surface $z = f(x, y)$ at the point $(a, b, f(a, b))$.
70. True or False: The function $f(x, y)$ has gradient ∇f at the point (a, b) . The vector $f_x(a, b)\mathbf{i} + f_y(a, b)\mathbf{j} + \mathbf{k}$ is perpendicular to the surface $z = f(x, y)$.
71. True or False: There exists a function f with continuous second-order partial derivatives such that $f_x(x, y) = x + y^2$ and $f_y(x, y) = x - y^2$.
72. True or False: $f_{xy} = \frac{\partial^2 f}{\partial x \partial y}$
73. True or False: $D_{\mathbf{k}}f(x, y, z) = f_z(x, y, z)$
74. True or False: If $f(x, y) \rightarrow L$ as $(x, y) \rightarrow (a, b)$ along every straight line through (a, b) , then $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$.
75. True or False: If $f_x(a, b)$ and $f_y(a, b)$ both exist, then f is differentiable at (a, b)
76. True or False: If f has a local minimum at (a, b) and f is differentiable at (a, b) , then $\nabla f(a, b) = \mathbf{0}$.
77. True or False: If f is a function, then

$$\lim_{(x,y) \rightarrow (2,5)} f(x, y) = f(2, 5).$$

78. True or False: If $f(x, y) = \ln(y)$, then $\nabla f(x, y) = 1/y$.
79. True or False: If $f(x, y)$ has two local maxima, then f must have a local minimum.
80. True or False: If $f(x, y) = \sin(x) + \sin(y)$, then $-\sqrt{2} \leq D_{\mathbf{u}}f(x, y) \leq \sqrt{2}$.

3 Multiple Integration, Polar, Cylindrical, Spherical Coordinates, Change of Variables

81. True or False: The cylindrical coordinates of the point with Cartesian coordinates $(4, \pi, 0)$ are $(-4, 0, 0)$.
82. True or False: The equation $r = 2 \sin(\theta)$ describes a circle.
83. True or False: The mass of a region in the plane might be given by $\int_0^x \int_0^y 3 \, dy \, dx$.
84. True or False: If $f(x, y) > 0$ for all $x, y \in \mathbb{R}^2$,

$$\int_0^1 \int_0^1 f(x, y) \, dy \, dx > \int_0^1 \int_x^1 f(x, y) \, dy \, dx.$$

85. True or False: The region $0 \leq \rho \leq 4, 0 \leq \phi \leq \pi, 0 \leq \theta \leq \pi$ describes the bottom half of a sphere of radius 4 centered on the origin.

86. True or False:

$$\int_{-1}^3 \int_{10}^{353,203} x^2 y^3 + y^{17} + (2-x)^{362} dx dy = \int_{10}^{353,203} \int_{-1}^3 x^2 y^3 + y^{17} + (2-x)^{362} dy dx$$

87. True or False: Suppose $f(x, y)$ is a continuous function and R is a region bounded by a smooth curve in the plane. If $\iint_R dA = 4$ and the average value of f on R is 3, then $\iint_R f(x, y) dA = \frac{3}{4}$.

88. True or False:

$$\int_{-2}^2 \int_0^{\sqrt{4-x^2}} \sqrt{x^2 + y^2} dy dx = \int_0^\pi \int_0^2 r dr d\theta.$$

89. True or False: Let E be an ellipse in $\mathbb{R}^2$. If $\iint_E 1 dA = 10$ and $\iint_E f(x, y) dA = 20$, then the average value of $f(x, y)$ on E is 2.

90. True or False:

$$\int_0^1 \int_0^x \sqrt{x+y^2} dy dx = \int_0^x \int_0^1 \sqrt{x+y^2} dx dy$$

91. True or False:

$$\int_1^2 \int_3^4 x^2 e^y dy dx = \int_1^2 x^2 dx \int_3^4 e^y dy$$

92. True or False:

$$\int_{-1}^1 \int_0^1 e^{x^2+y^2} \sin(y) dx dy = 0$$

93. True or False: If $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous on $[0, 1]$, then

$$\int_0^1 \int_0^1 f(x)f(y) dy dx = \left(\int_0^1 f(x) dx \right)^2$$

94. True or False:

$$\int_1^4 \int_0^1 (x^2 + \sqrt{y}) \sin(x^2 y^2) dx dy \leq 9$$

95. True or False: The integral

$$\int_0^{2\pi} \int_0^2 \int_r^2 dz dr d\theta$$

represents the volume enclosed by the cone $z = \sqrt{x^2 + y^2}$ and the plane $z = 2$.

96. True or False: If R is the union of the disjoint regions R_1 and R_2 in $\mathbb{R}^2$, then

$$\iint_R f(x, y) dA = \iint_{R_1} f(x, y) dA - \iint_{R_2} f(x, y) dA.$$

4 Vector Fields, Line Integrals, Surface Integrals, Conservative Vector Fields, Stokes' Theorem

97. True or False: Every vector field $\mathbf{F}(x, y)$ is a gradient vector field, i.e. there is always some $f(x, y)$ so that $\mathbf{F} = \nabla f$.
98. True or False: If $\mathbf{F}(x, y, z)$ is a velocity field for a fluid flowing in space, then $\nabla \cdot \mathbf{F}$ is a vector whose direction is the right-hand rule direction of the axis of rotation of the fluid at each point.
99. True or False: For any continuously differentiable function $f(x, y, z)$, $\nabla \times (\nabla f) = \mathbf{0}$.
100. True or False: If C_1 and C_2 are two smooth paths from $(0, 0, 0)$ to $(1, 0, 1)$ and $f(x, y, z)$ is a differentiable function, then we must have

$$\int_{C_1} f(x, y, z) \, ds = \int_{C_2} f(x, y, z) \, ds.$$

101. True or False: If $\mathbf{F}(x, y)$ is a conservative vector field defined on all of $\mathbb{R}^2$ and C is any closed curve in the plane, then

$$\oint_C (\mathbf{F} \cdot \mathbf{T}) \, ds = 0.$$

102. True or False: Two different curves can be flow lines for the same vector field.
103. True or False: Given two circles centered at the origin, oriented counterclockwise, and any vector field $\mathbf{F}$, then the line integral of $\mathbf{F}$ is larger around the circle with larger radius.
104. True or False: If $\mathbf{F}$ is any vector field and C is a circle, then the integral of $\mathbf{F}$ around C traversed clockwise is the negative of the integral of $\mathbf{F}$ around C traversed counterclockwise.
105. True or False: If all the flow lines of a vector field $\mathbf{F}$ are parallel straight lines, then $\nabla \cdot \mathbf{F} = 0$.
106. True or False: If all the flow lines of a vector field $\mathbf{F}$ radiate outward along straight lines from the origin, then $\nabla \cdot \mathbf{F} > 0$.
107. True or False: The vector field $\mathbf{F}$ is defined everywhere in a region W bounded by a surface S . If $\nabla \cdot \mathbf{F} > 0$ at all points of W , then the vector field $\mathbf{F}$ points outward at all points of S .
108. True or False: The vector field $\mathbf{F}$ is defined everywhere in a region W bounded by a surface S . If $\nabla \cdot \mathbf{F} > 0$ at all points of W , then the vector field $\mathbf{F}$ points outward at some points of S .
109. True or False: The vector field $\mathbf{F}$ is defined everywhere in a region W bounded by a surface S . If $\iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma > 0$, then $\nabla \cdot \mathbf{F} > 0$ at some points of W .
110. True or False: If all the flow lines of a vector field $\mathbf{F}$ are straight lines, then $\nabla \times \mathbf{F} = \mathbf{0}$.
111. True or False: If all the flow lines of a vector field $\mathbf{F}$ lie in planes parallel to the xy -plane, then the curl of $\mathbf{F}$ is a multiple of $\mathbf{k}$ at every point.
112. True or False: If $\mathbf{F}$ is a vector field, then $\nabla \cdot \mathbf{F}$ is a vector field.
113. True or False: If $\mathbf{F}$ is a vector field, then $\nabla \times \mathbf{F}$ is a vector field.
114. True or False: If f has continuous partial derivatives of all orders on $\mathbb{R}^3$, then $\nabla \cdot (\nabla \times (\nabla f)) = 0$.
115. True or False: If f has continuous partial derivatives on $\mathbb{R}^3$ and C is any circle, then $\int_C \nabla f \cdot d\mathbf{r} = 0$.
116. True or False: If $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$ and $P_y = Q_x$ in an open region D , then $\mathbf{F}$ is conservative.

117. True or False: If $\mathbf{F}$ and $\mathbf{G}$ are vector fields and $\nabla \cdot \mathbf{F} = \nabla \cdot \mathbf{G}$, then $\mathbf{F} = \mathbf{G}$.
118. True or False: The work done by a conservative force field in moving a particle around a closed path is zero.
119. True or False: If $\mathbf{F}$ and $\mathbf{G}$ are vector fields, then $\nabla \times (\mathbf{F} + \mathbf{G}) = \nabla \times \mathbf{F} + \nabla \times \mathbf{G}$.
120. True or False: If $\mathbf{F}$ and $\mathbf{G}$ are vector fields, then $\nabla \times (\mathbf{F} \cdot \mathbf{G}) = (\nabla \times \mathbf{F}) \cdot (\nabla \times \mathbf{G})$.
121. True or False: If S is a sphere in $\mathbb{R}^3$ and $\mathbf{F}$ is a constant vector field, then $\iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma = 0$.
122. True or False: There is a vector field $\mathbf{F}$ such that $\nabla \times \mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.
123. True or False: Suppose S is a smooth surface in $\mathbb{R}^3$ and $\mathbf{F}$ is a vector field in $\mathbb{R}^3$. If $\iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma > 0$, then the angle between $\mathbf{F}$ and $\mathbf{n}$ is acute at all points on S .